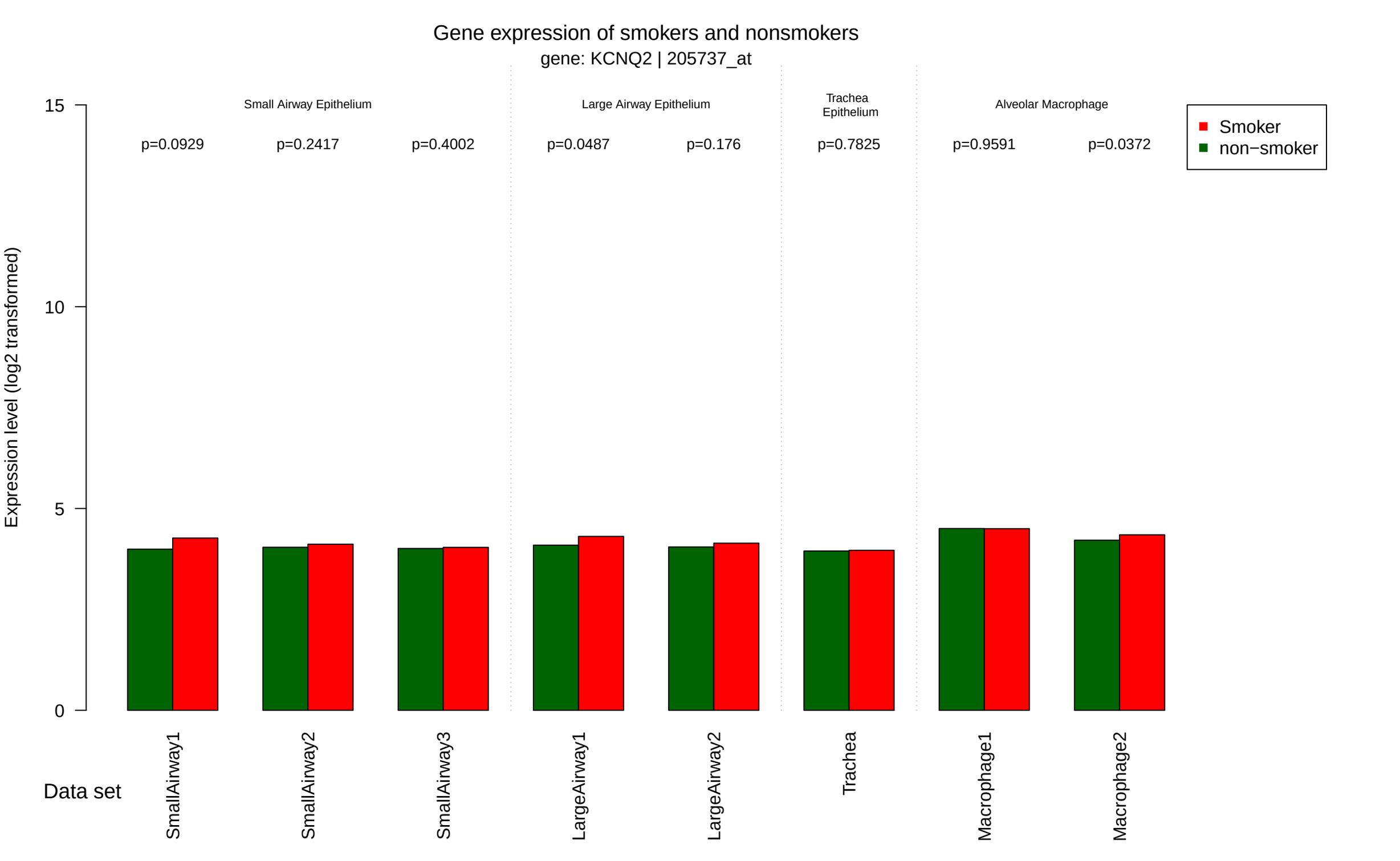
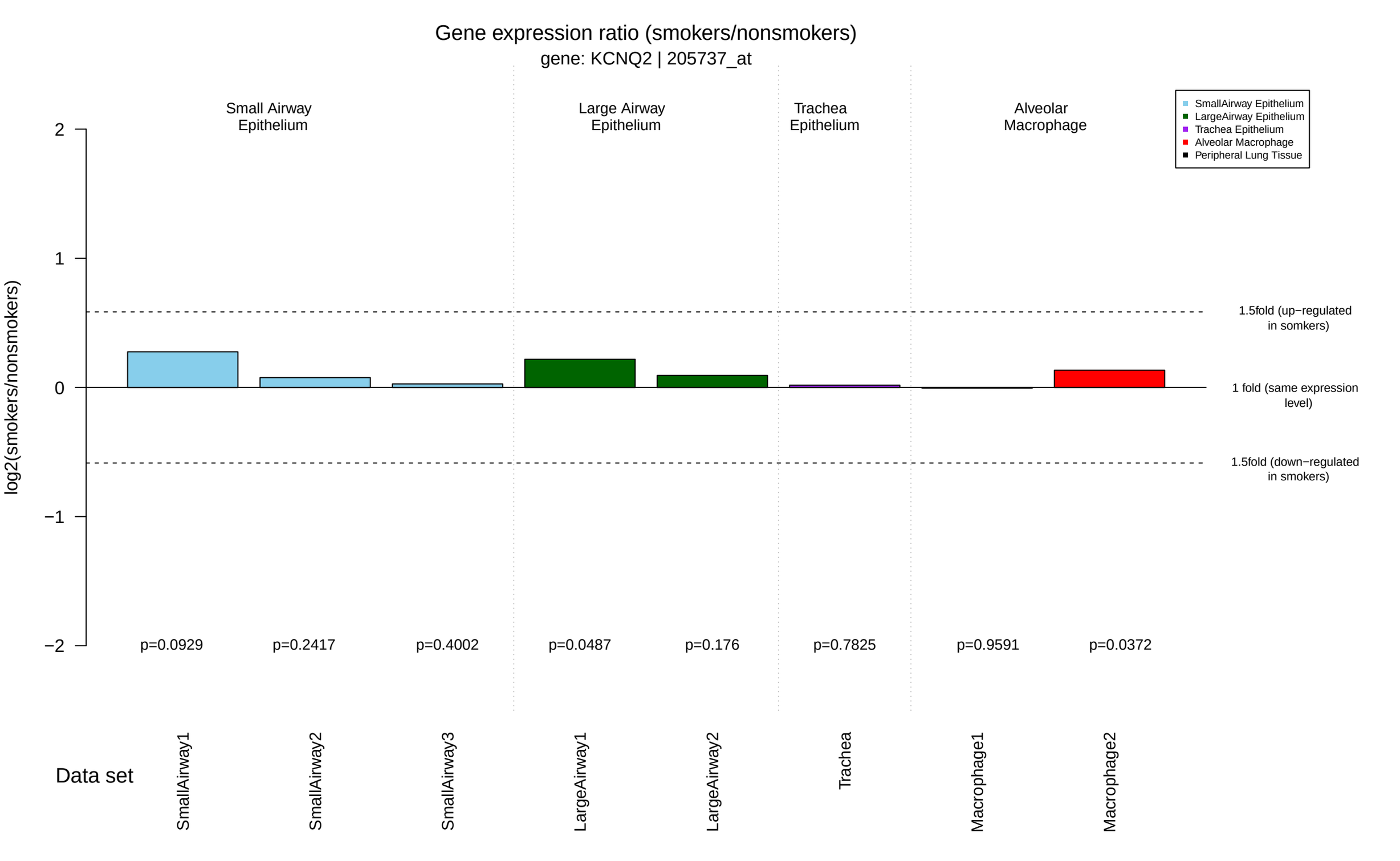
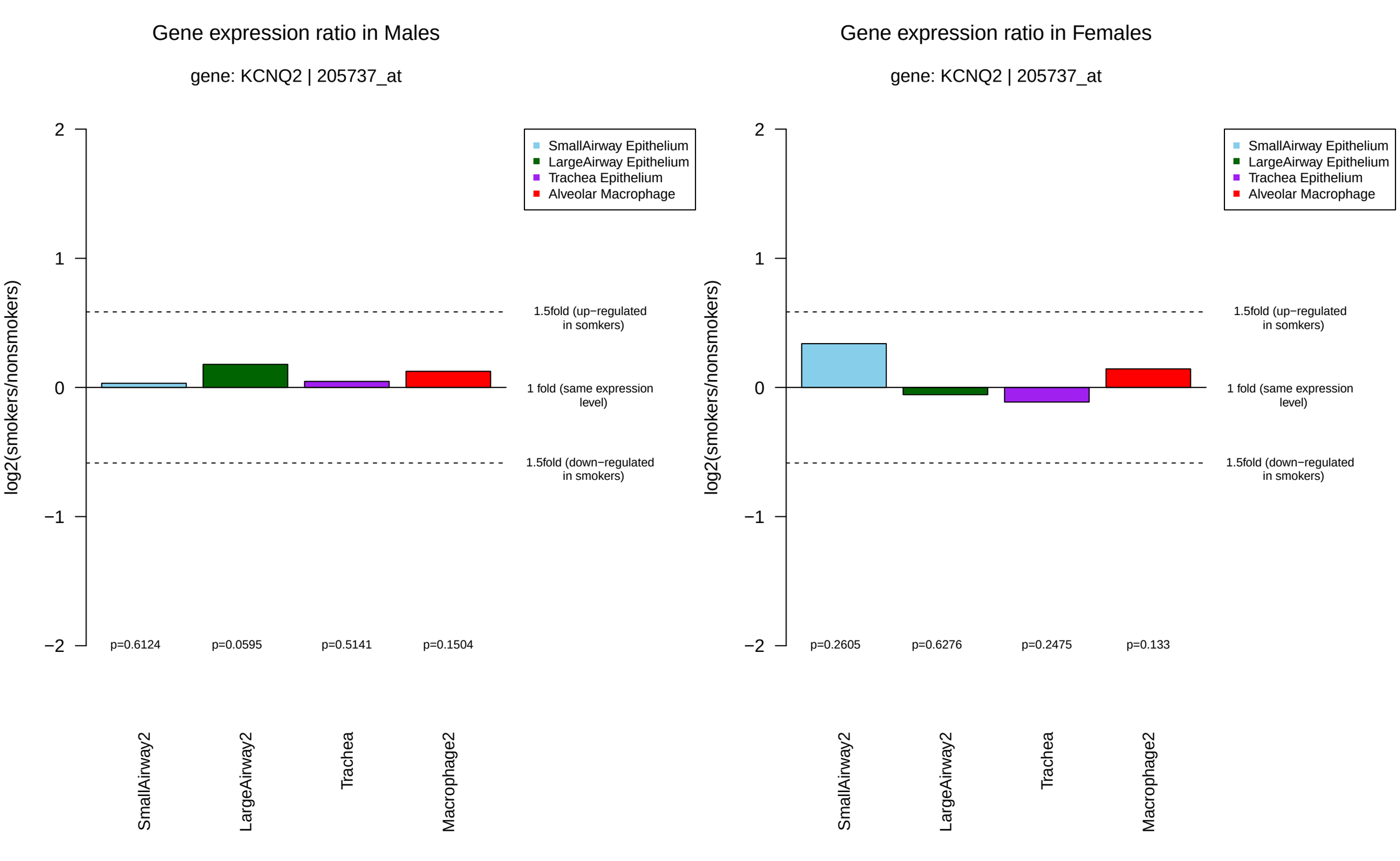
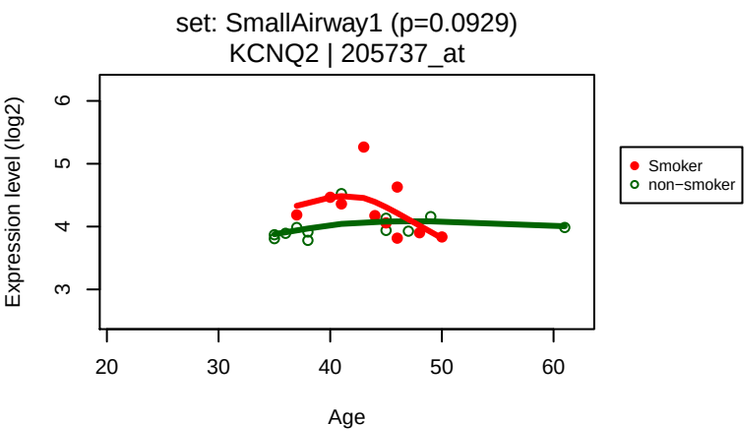
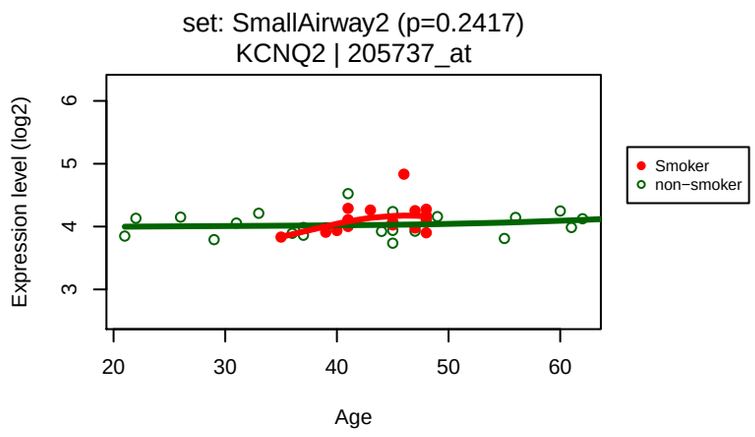
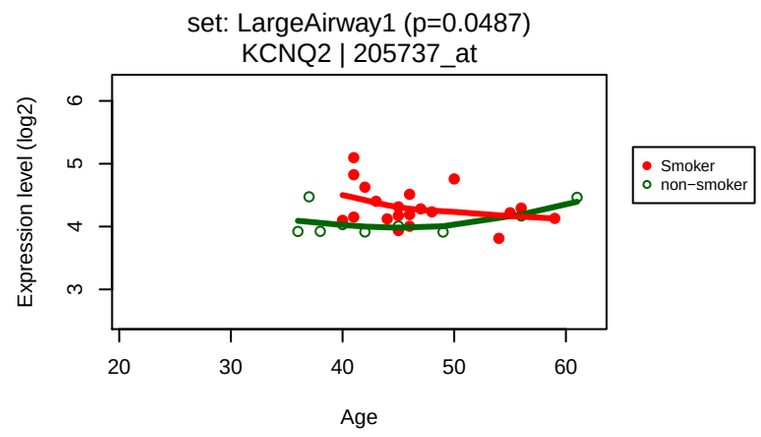

Gene expression of smokers and nonsmokers
gene: KCNQ2 | 205737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
p=0.0929
p=0.2417
p=0.4002
p=0.0487
p=0.176
p=0.7825
p=0.9591
p=0.0372
Smoker
non−smoker
Expression level (log2 transformed)
15
10
5
0
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

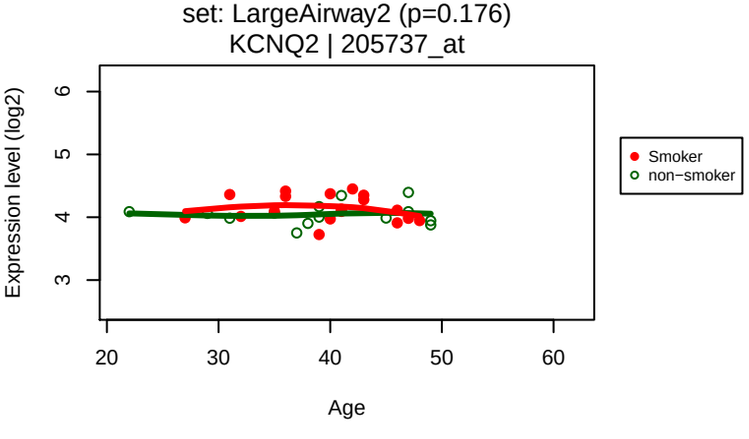

Gene expression ratio (smokers/nonsmokers)
gene: KCNQ2 | 205737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Peripheral Lung Tissue
log2(smokers/nonsmokers)
2
1
0
−1
−2
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.0929
p=0.2417
p=0.4002
p=0.0487
p=0.176
p=0.7825
p=0.9591
p=0.0372
Data set
SmallAirway1
SmallAirway2
SmallAirway3
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2

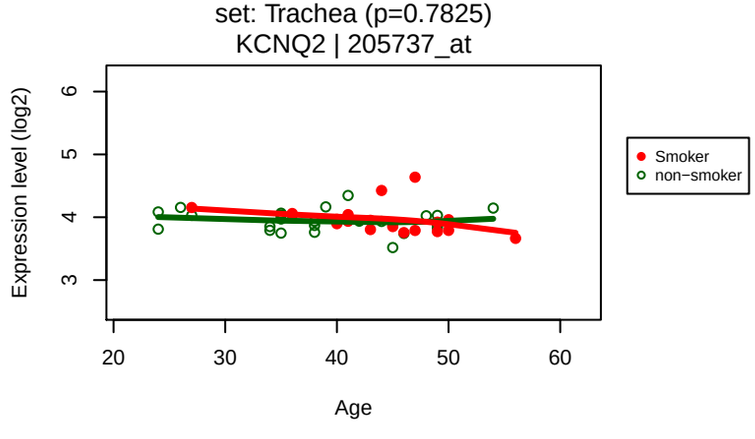

Gene expression ratio in Males
gene: KCNQ2 | 205737_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.6124
p=0.0595
p=0.5141
p=0.1504
SmallAirway2
LargeAirway2
Trachea
Macrophage2
Gene expression ratio in Females
gene: KCNQ2 | 205737_at
log2(smokers/nonsmokers)
2
1
0
−1
−2
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
1.5fold (up−regulated in somkers)
1 fold (same expression level)
1.5fold (down−regulated in smokers)
p=0.2605
p=0.6276
p=0.2475
p=0.133
SmallAirway2
LargeAirway2
Trachea
Macrophage2

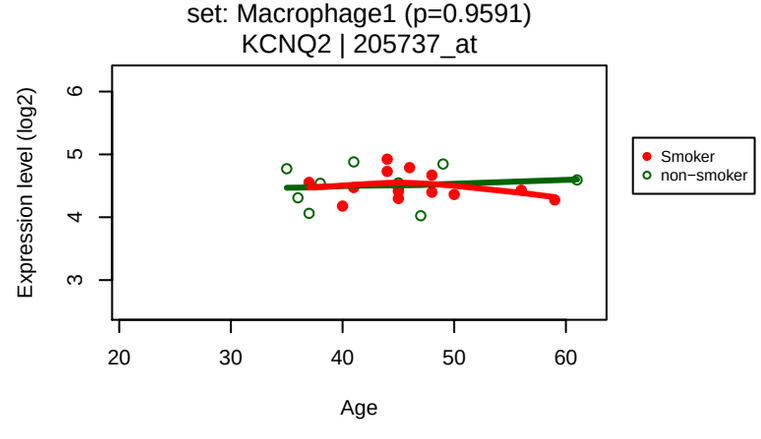

set: SmallAirway1 (p=0.0929)
KCNQ2 | 205737_at
Expression level (log2)
Age
Smoker
non−smoker

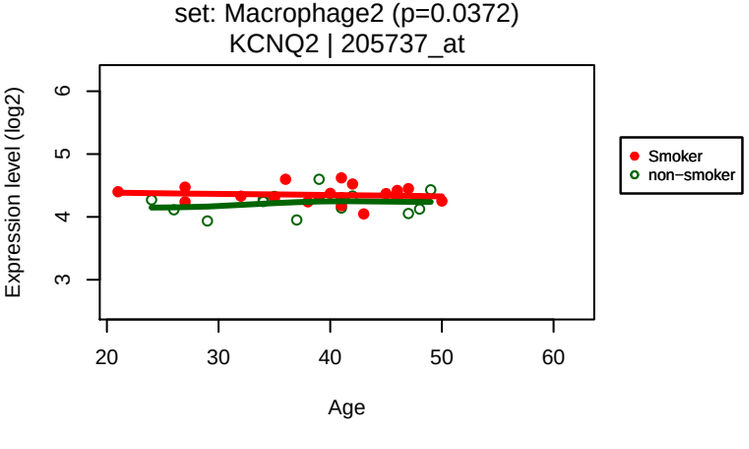

set: SmallAirway2 (p=0.2417)
KCNQ2 | 205737_at
Expression level (log2)
Age
Smoker
non−smoker

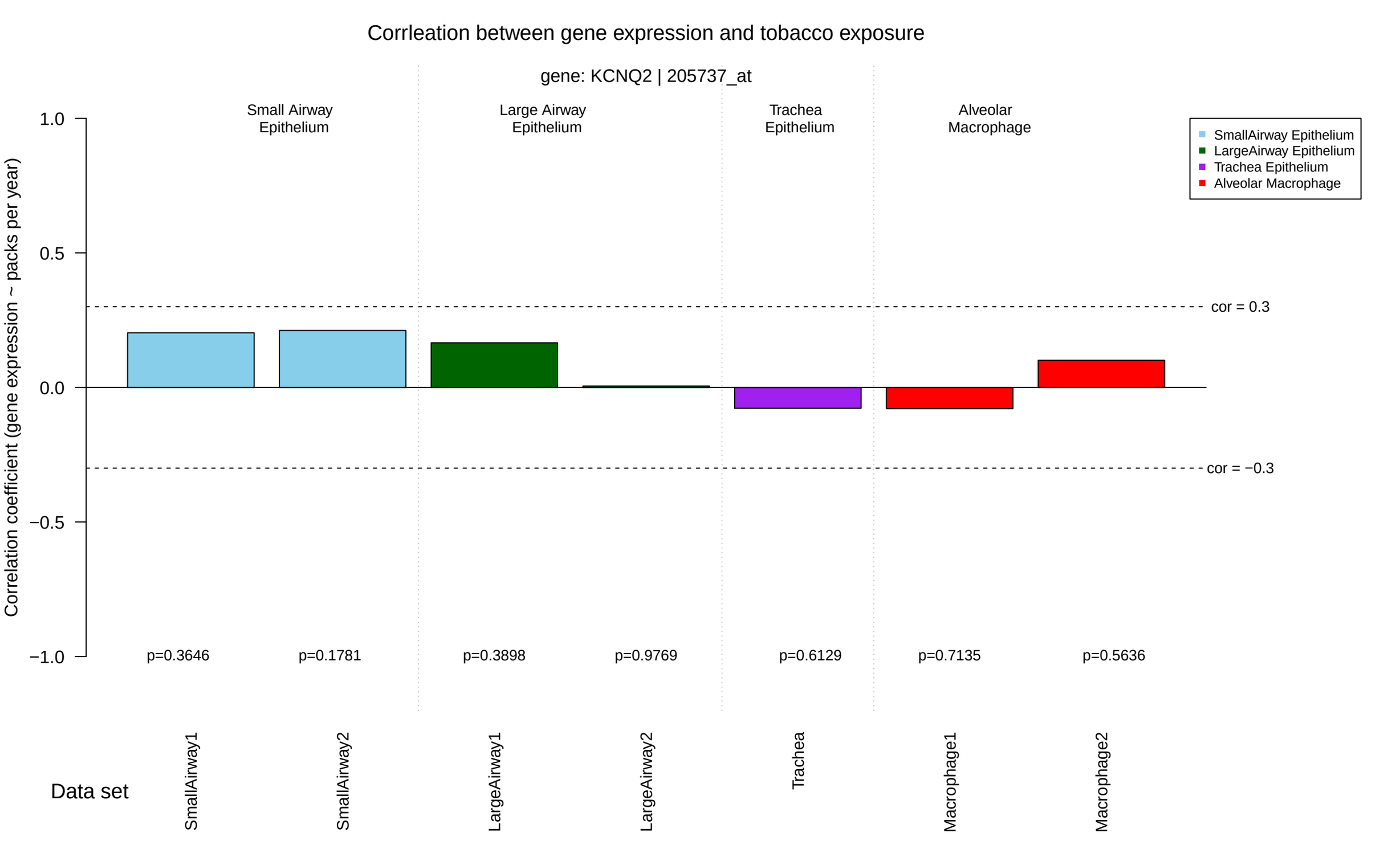

set: LargeAirway1 (p=0.0487)
KCNQ2 | 205737_at
Expression level (log2)
Age
Smoker
non−smoker


set: LargeAirway2 (p=0.176)
KCNQ2 | 205737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Trachea (p=0.7825)
KCNQ2 | 205737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage1 (p=0.9591)
KCNQ2 | 205737_at
Expression level (log2)
Age
Smoker
non−smoker


set: Macrophage2 (p=0.0372)
KCNQ2 | 205737_at
Expression level (log2)
Age
Smoker
non−smoker


Corrleation between gene expression and tobacco exposure
gene: KCNQ2 | 205737_at
Small Airway Epithelium
Large Airway Epithelium
Trachea Epithelium
Alveolar Macrophage
SmallAirway Epithelium
LargeAirway Epithelium
Trachea Epithelium
Alveolar Macrophage
Correlation coefficient (gene expression ~ packs per year)
1.0
0.5
0.0
−0.5
−1.0
cor = 0.3
cor = −0.3
p=0.3646
p=0.1781
p=0.3898
p=0.9769
p=0.6129
p=0.7135
p=0.5636
Data set
SmallAirway1
SmallAirway2
LargeAirway1
LargeAirway2
Trachea
Macrophage1
Macrophage2